

The background is a dark blue-grey gradient. On the left, there are two overlapping geometric shapes: a blue parallelogram and a light green parallelogram. Below these, a circular inset shows a detailed, high-magnification view of a printed circuit board (PCB) with various electronic components. In the top right corner, there is a faint, grey, 3D-rendered pattern resembling a complex circuit or a topographical map.

Econometrics Group project

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Data sources and libraries used

All the relevant data have been extracted from the [World Bank open data platform](#). In particular the following databases have been used:

- [GDP per capita \(constant 2015 US\\$\)](#) - representing the GDP per capita of countries, all currencies converted to the 2015 U.S. dollar;
- [Population growth \(annual %\)](#) - representing annual population growth rate (counting all residents regardless of legal status/citizenship);
- [Net savings](#) - representing Gross domestic savings, that is GDP less final consumption expenditure (represented as % of GDP);
- [Net migration](#) - representing net total of migrants during a period, that is, the number of immigrants minus the number of emigrants, including both citizens and noncitizens. The variable net migration has been chosen to improve the regression since it's reasonable to assume that a country with an high positive net migration represents a country in which working opportunities are favorable and the economy is healthy and growing (thus the country will present a high GDP per capita). An analogue (but opposite) reasoning can be applied in case of negative net migration.

All the data used span from 1961 (included) to 2022 (included).

In order to deal with missing observations and anomalies in the dataset, indicators for groups of countries have been removed and two subsets of countries have been created: OECD and developing countries (see code for list). (Note: for OECD countries we included all countries into the OECD list before 1991). To improve the quality of our input data, we selected a subsample of developing countries which had complete observations across the time span (defined in the code as "dev_countries_best").

To handle the net migration variable, which can take up also negative values, we performed the computation of the logarithm of the absolute value. The transformed values accurately reflect the magnitude and direction of change experienced by each country.

All the regression have been performed using the logarithm of variables, thus variable coefficients represent the elasticity between the explanatory variable and the dependent variable.

Testing the Solow-Swan growth model: OECD Countries

Assumptions

Assuming that OECD countries reported data with higher quality with respect to all the countries present in the database, we decided to first test the Solow-Swan model on OECD countries only.

Interpretation of results:

The solow model performs poorly in explaining the GDPpc of OECD countries, this may be for various reasons, such as:

1. OECD countries are closer to their Steady state;
2. OECD countries variables move similarly;
3. There is lower variance in explanatory variables of OECD countries.

OLS Assumption Testing

- **Reset Test:** The model pass the reset test, thus the parameters are well specified and parameters are in a linear relationship.
- **Breusch-Pagan:** The test rejects the null hypothesis of homoscedasticity. Heteroskedasticity may be another reason for the poor results of this regression
- **DW:** : The model doesn't pass the DW test, this is something we expected since we are considering data from a time series
- **JB:** The model pass the Jarque-Bera test, thus errors are normally distributed, the p-value is however quite low compared to the one of the other regressions

Table 1: Results

	Dependent variable:		
	GDPpc-OECD (1)	GDPpc-dev (2)	GDPpc-dev-best (3)
savings-OECD	2.05478** (0.94217)		
growth-OECD	-0.75025*** (0.19606)		
savings-dev		1.41153*** (0.21379)	
growth-dev		-0.93004*** (0.09953)	
savings-dev-best			0.68202*** (0.21613)
growth-dev-best			-1.12024*** (0.07547)
Constant	3.41191 (3.08676)	4.66316*** (0.64952)	6.91120*** (0.67947)
Observations	62	62	62
R ²	0.20662	0.74926	0.91059
Adjusted R ²	0.17972	0.74076	0.90756
Residual Std. Error (df = 59)	0.35965	0.16741	0.09579
F Statistic (df = 2; 59)	7.68246***	88.15150***	300.43650***

Testing the Solow-Swan growth model: Developing countries

Assumptions

We thus decided to test the model with developing countries, since these regions present more variability in the data and higher economic growth in the period taken into consideration. The [list of developing countries](#), was subsequently reduced to exclude those who presented incomplete data for the period taken into consideration

Interpretation of results:

The solow model performs way better for developing countries and even better for the restricted dataset of countries with complete observations. The latter obtained an R^2 higher than 0.9, testifying the high performance of the model in describing the variation in GDPpc. Considering the latter, a 1% increase in savings is associated with a 0.6% increase in GDPpc and a 1% increase in population growth is associated with a 1.12% decrease in GDPpc.

OLS Assumption Testing

- **Reset Test:** The model pass the reset test, thus the parameters are well specified and parameters are in a linear relationship.
- **Breusch-Pagan:** The model successfully pass the test, thus there is Homoscedasticity in the residuals distribution
- **DW:** The model doesn't pass the DW test, this is something we expected since we are considering data from a time series
- **JB:** The model pass the Jarque-Bera test, thus errors are normally distributed

Test	Results: OECD countries
RESET test:	RESET = 3.5198, df1 = 2, df2 = 57, p-value = 0.3619
Breusch-Pagan test:	BP = 18.864, df = 2, p-value = 8.012e-05
Durbin-Watson test:	DW = 0.133393, p-value < 2.2e-16
Jarque-Bera test:	X^2 = 4.7552, df = 2, p-value = 0.09277

Test	Results: developing countries best
RESET test:	RESET = 0.6752, df1 = 2, df2 = 57, p-value = 0.5131
Breusch-Pagan test:	BP = 1.574, df = 2, p-value = 0.4552
Durbin-Watson test:	DW = 0.23534, p-value < 2.2e-16
Jarque-Bera test:	X^2 = 1.2102, df = 2, p-value = 0.546

Test	Results: developing countries best with migration
RESET test:	RESET = 0.60363, df1 = 3, df2 = 56, p-value = 0.5503
Breusch-Pagan test:	BP = 7.6286, df = 3, p-value = 0.05434
Durbin-Watson test:	DW = 0.3076, p-value = 0.0
Jarque-Bera test:	X^2 = 0.67589, df = 3, p-value = 0.7132



Expanding the Solow-Swan growth model

Assumptions

We decided to expand the Solow-Swan model by investigating whether a country's net migration may have an influence on the nation GDPpc. We focused our analysis on the developing countries presenting complete data, since they were the ones to report better regression results in the previous stages of our research. We sustain that incorporating the net migration variable may contribute in better estimating GDPpc, as a positive net migration signifies attractiveness to workers and is reflected in increased labor availability (vice versa being valid in case of negative migration).

Interpretation of results

The extended model works greatly at explaining GDPpc, indeed including net migration improves the result of our regression, in fact both R^2 and adjusted R^2 are higher, thus the improvement is not due only to the fact that we added a new variable, but the variable is indeed significative. As we assumed there is a positive relationship between Net-Migration and GDPpc

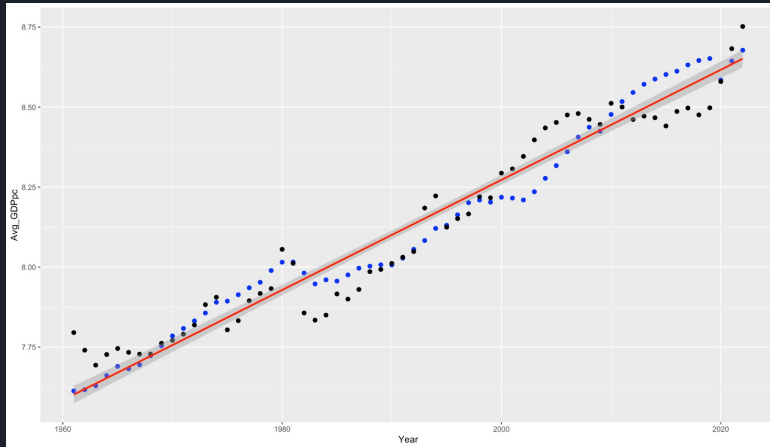
OLS Assumption Testing

- **Reset Test:** Now the model pass the reset Test, demonstrating that it's better formulated
- **Breusch-Pagan:** The results of the BP test are slightly worse compared to the prior model but still acceptable at a 95% confidence level
- **DW:** The model still doesn't pass the DW test, clearly adding a new variable doesn't solve the problem that errors are serially correlated in a time series
- **JB:** The model pass the Jarque-Bera test, errors are still normally distributed.

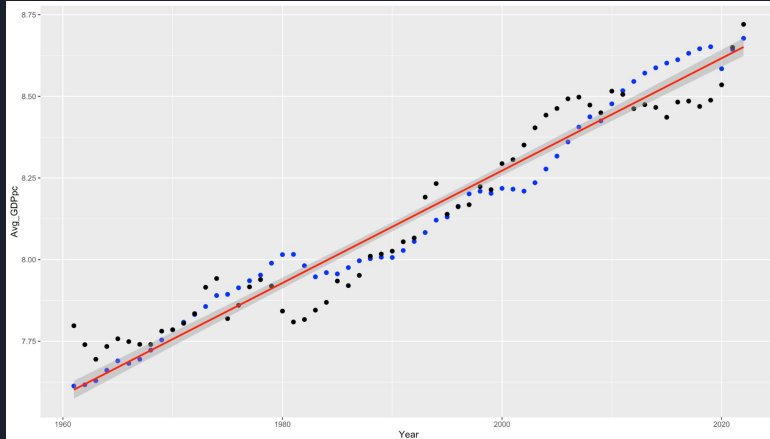
Table 2: Results with Migration

	<i>Dependent variable:</i>
	GDPpc-dev-best-migration
savings-dev-best	0.568*** (0.198)
growth-dev-best	-1.217*** (0.073)
net-migration-dev-best	0.027*** (0.007)
Constant	7.433*** (0.632)
Observations	62
R ²	0.928
Adjusted R ²	0.924
Residual Std. Error	0.087 (df = 58)
F Statistic	248.463*** (df = 3; 58)

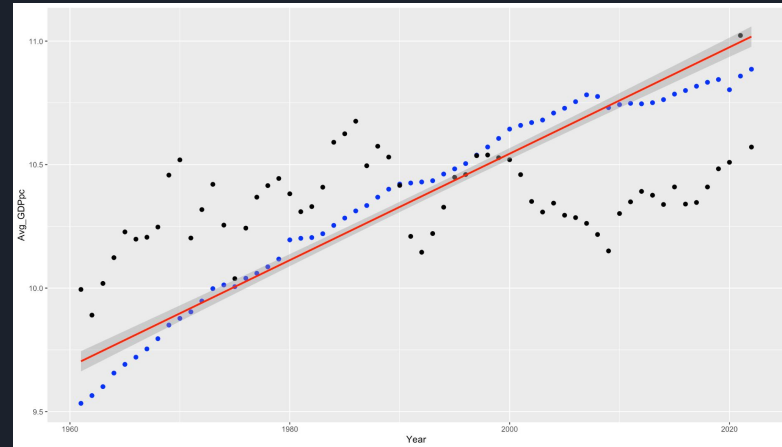
Plots of the regressions



Regression on OECD countries



Regression on selected developing countries



Regression on selected countries including net migration

Graphical interpretation

Plotting the results of the regression highlights clearly how far from the actual values where the predicted values in the regression with OECD countries.

The results of the regression on the selected developing countries and the one including net migration look quite similar, however it is noticeable that including the net migration variable helps explaining better the values between 1970 and 1990.

* On the y axis avg_GDPpc is plotted in a logarithmic scale.

** The blue dots represent the true values of GDPpc while the black dots represent the predicted values